PV102 Day 2 Student Packet

Site and Solar Resource Assessment

Today's rule

Measure the site before you size the system.

Learning targets

- Identify site factors that affect PV production and installation practicality.
- Use orientation, tilt, shade, and access observations to support a design decision.
- Separate measured observations from assumptions and preferences.
- Create a site-note handoff that Day 3 can use for load and energy work.

Morning quiz

1. Which site condition most directly reduces solar access?

- A. Good labeling
- B. Shade on the array area
- C. A clean classroom
- D. A larger clipboard

2. Azimuth describes:

- A. Direction/orientation
- B. Battery chemistry
- C. Wire color
- D. Breaker rating

3. Tilt describes:

- A. Array angle relative to horizontal
- B. Fuse speed only
- C. Type of current
- D. Wire insulation color

4. A site note should separate observations from:

- A. Measurements
- B. Evidence
- C. Assumptions
- D. Safety boundaries

5. The safest first action before a roof assessment is to:

- A. Climb quickly
- B. Discuss access, fall hazards, and boundaries
- C. Carry modules first
- D. Energize the inverter

6. Name one factor besides shade that affects PV placement.

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7. A team records "probably enough sun." What should they do next?

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8. Why can a sunny site still be a poor installation site?

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9. What is one unknown that should be carried into Day 3?

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10. Write the Day 2 rule in your own words.

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Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Site Walk and Solar Access Handoff

Teams inspect three candidate locations and decide which one deserves further design work.

Scenario options

- South-facing roof plane with one morning tree shadow and limited attic access.
- Open ground rack area with excellent sun but long wire run and mower traffic.
- West-facing shed roof near the load with easy service access and afternoon sun.

Team procedure

- Assign one candidate location per team.
- Sketch the site from above and mark north, shade sources, access path, approximate array area, and likely wire route.
- Record observations in three columns: measured, observed, unknown.
- Name the best reason to keep or reject the site for first-pass design.
- Prepare a two-minute handoff for Day 3.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which site condition would most likely force a redesign, and what evidence supports that answer?

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